MIL-D-12150E

22 November 1976

SUPERSEDING
MIL-D-12150D
29 December 1967

MILITARY SPECIFICATION

DETERGENT, DRY CLEANING SOLVENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type and grade of dry cleaning detergent (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-T-236	-	Tetrachloroethylene (Perchloroethylene), Technical
P-D-680	-	Dry Cleaning Solvent
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28-Gage and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U. S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7930

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STANDARDS

FEDERAL

FED-STD-536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing
FED-STD-791 - Lubricants, Liquid Fuels, and Related Products: Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

16 CFR 1500 - Federal Hazardous Substance Act

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD

E70 - Standard Method of Test for pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical societies and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. The dry cleaning detergent shall consist of a minimum of 94 percent active detergent plus a liquid nonionic surface active agent. The mixture shall conform to the requirements of table I, when tested as specified in 4.3.2.

TABLE I. Chemical and Physical Requirements

Property	Requirement
Homogeneity	No phase separation
Odor:	
Initial	Not acrid or rancid
Residual	Not rancid
Flashpoint, minimum	175°F (79°C)
pH value	6.0 - 9.0
Solubility:	
Drycleaning solvent	No separation
Tetrachloroethylene	No separation
Moisture, percent maximum	6.0
Soil suspension value, minimum	1.0
Biodegradability, percent minimum	90.0

3.2 Labeling. The following labeling in compliance with the provisions of the Federal Hazardous Substances Labeling Act shall appear on all containers:

CAUTION - COMBUSTIBLE LIQUID

THIS PRODUCT IS COMBUSTIBLE AND MUST NOT BE USED OR STORED IN AREAS WHERE THERE ARE OPEN FLAMES, ARCING EQUIPMENT, HOT SURFACES OR WHERE SMOKING IS PERMITTED. AVOID INHALING VAPORS. DO NOT GET IN EYES OR ON SKIN. IN CASE OF CONTACT, IMMEDIATELY FLUSH EYES OR SKIN WITH PLENTY OF WATER FOR AT LEAST 15 MINUTES: FOR EYES, GET PROMPT MEDICAL ATTENTION.

3.3 Workmanship. The detergent shall be free of foreign matter, and shall not phase separate.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions of MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examinations and tests shall consist of all the end items prepared in a single batch and submitted for inspection at one time.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs below at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for packaging inspection under 4.3.1.3. The end item lot shall consist of dry cleaning detergent manufactured in one batch.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one filled unit container.

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Examine	Defect
Construction	Not as specified (see 5.1). Container broken, chipped, dirty, any leaking of contents. Any break or severe dent, causing sharp ridge or buckling in body of container. Not in accordance with contract requirements.
Closure	Not type specified; imperfect.
Seal	Not type specified.
Labeling	Missing, illegible. Does not contain labeling information specified in 3.2. Not as specified (see 5.3).
Contents	Not clear liquid. Contains foreign matter.

4.3.1.2 Examination of the end item for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net content per container is less than specified.

* 4.3.1.3 Packaging inspection. Packaging inspection shall be in accordance with the applicable requirements in the appendix of PPP-C-96.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A

* 4.3.2 Testing of the end item. The end item shall be tested for the characteristics specified in table I. The number of sample units for testing shall be as specified in 4.3.2.1 and 4.3.2.2. The lot size for test purposes shall be the number of gallons of detergent submitted for inspection at one time. The sample size shall be shown below. When specified in table II, tests shall be performed on a composite sample or sample unit basis. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any test requirement specified.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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* 4.3.2.1 Composite sample. The sample unit for composite testing shall be 1 pint of detergent selected in equal portions from each sample unit. The portions from the individual samples shall be thoroughly mixed, yielding the composite. The composite sample unit shall be placed in an airtight and watertight glass container with suitable caps, covered and sealed to prevent any moisture change.

* 4.3.2.2 Sample unit. The sample unit for testing on a sample unit basis shall be 1/2 pint placed in an airtight and watertight glass container with suitable caps, covered and sealed to prevent any moisture change.

* TABLE II. Instructions for testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number of determinations per sample unit	Results reported as	
	Requirement	Test method	Individual unit	Composite sample		Pass or fail 1/	Numerically to nearest
Homogeneity	Table I	4.4.1		X	1	X	
Odor	Table I	4.4.2					
Initial				X	1	X	
Residual				X	1	X	
Flash point, F° (C°)	Table I	4.4.3		X	Average of 2		1.0°F (0.5°C)
pH value	Table I	4.4.4		X	Average of 2		0.1 unit
Solubility	Table I	4.4.5					
Drycleaning solvent				X	1	X	
Tetrachloroethylene				X	1	X	
Moisture content	Table I	4.4.6	X		Average of 2		0.1 percent
Soil suspension value	Table I	4.4.7		X	1		0.1 unit
Synthetic organic detergent	3.1	2/					
Biodegradability, percent	Table I	4.4.8					

1/ If failure is indicated, report description or numerical point of failure, as applicable.

2/ The contractor shall furnish a certificate of compliance stating the chemical nature of the organic synthetic detergent and that the detergent concentration in the product as furnished is at least 94 percent.

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4.4 Test procedures.

* 4.4.1 Appearance. Thoroughly mix the sample, pour into a 250 ml. graduate cylinder, allow the sample to stand at room temperature for 24 hours and check for appearance. The sample shall show no phase separation and shall be homogeneous except that it may contain a small amount of flocculent precipitate that can be readily dispersed by agitation.

4.4.2 Odor.

4.4.2.1 Initial. The odor of the sample shall not be rancid or acrid.

* 4.4.2.2 Residual. Allow a 4-inch by 4-inch (10 cm by 10 cm) swatch of white wool cloth (see 6.4) to stand in a 0.75 percent by volume solution of the detergent in type I solvent specified in P-D-680 for 30 minutes. Remove the swatch without wringing, allow to drain on a wire screen for two minutes, place in a wide mouth bottle and shake with 200 ml. of the solvent for 5 minutes. Repeat the rinse for another 5 minutes using fresh solvent, and then hang the swatch to dry overnight at room temperature. Place in an oven at 100° ± 3°F (38 ± 1°C) for 2 hours. The sample shall be examined for rancid odors before and after placing in the oven. Rancid odor present after oven drying shall constitute failure.

4.4.3 Flash point (Pensky-Martens). Flash point shall be determined in accordance with Method 1102 of FED-STD-791.

4.4.4 pH value. Prepare a 10 percent emulsion using 10 ml. of dry cleaning detergent and 90 ml. of freshly boiled and cooled distilled water. Determine the acidity of this emulsion in accordance with ASTM Designation E70.

* 4.4.5 Solubility. Dilute 4 ml. dry cleaning detergent to a volume of 96 ml. using type I solvent specified in P-D-680. Repeat the same procedure but using tetrachloroethylene conforming to Q-T-236. These solutions shall be examined after two hours. Any evidence of separation in either solution shall constitute failure.

* 4.4.6 Moisture content. The moisture content shall be determined in accordance with Method 202 of FED-STD-536.

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* 4.4.7 Soil suspension value. Condition eight 4-inch by 4-inch (10 cm by 10 cm) swatches of white wool fabric (see 6.4), for 48 hours in a desiccator over a saturated solution of Ca $(NO_3)_2$ at room temperature. Place 400 ml. of type I dry cleaning solvent as specified in P-D-680 containing 0.5 percent by volume of the dry cleaning detergent under test in each of two 1-pint (0.473 1) Mason jars. Into each of two other 1-pint (0.473 1) Mason jars introduce 400 ml. of type I dry cleaning solvent as specified in P-D-680 containing no dry cleaning detergent. Into each of the four jars add 0.200 gram of L. Martin's Globe Brand Carbon Black, or equal (see 6.3) and disperse for 3 minutes with a high speed stirrer. Place all four jars in a Launder-Ometer at room temperature and operate for 10 minutes at 41 $\pm$ 1 r.p.m. Introduce two white conditioned swatches into each jar and operate Launder-Ometer for an additional 20 minutes at room temperature. Remove the swatches from the jars, place upon a screen to drain and air dry on paper towels. Measure the light reflectance on each side of the swatches with a Hunter reflectometer or equivalent instrument and average the readings.

4.4.7.1 Calculation. Calculate the soil suspension value as follows:

$$\text{Soil suspension value} = \frac{\text{Average reflectance of fabric using detergent}}{\text{Average reflectance of fabric using no detergent}}$$

4.4.8 Biodegradability. Until an acceptable method of determining the biodegradability of a non-ionic detergent is agreed upon by industry and is found acceptable to the Government, the contractor shall furnish a certificate of compliance certifying that his product meets the generally accepted criteria for degree of degradation in accordance with a current commercial practice. The certificate shall be accompanied by actual test data (field or laboratory report) indicating the current commercial practice used in making the degradation determination.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A. Detergent shall be preserved-packaged in a 1-gallon (3.79 1) oblong cans with double-seamed on ends, conforming to type V, class 4 of PPP-C-96. Each can shall be fabricated from 1.25-pound hot-dipped tinplate or 0.50-pound electrolytic tinplate. Each can shall be exterior coated in accordance with plan B. When specified (see 6.2), each can shall be provided with a bridge type handle. Closure of each can shall be in accordance with the appendix of PPP-C-96.

5.1.2 Level C. Detergent shall be preserved-packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A packing. Detergent, preserved-packaged as apecified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

* 5.2.2 Level B packing. Detergent, preserved-packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

* 5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2) the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.2.3 Level C packing. Detergent, preserved-packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-C-96.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each can shall marked with the caution information specified in 3.2 by legibly printing or stamping in black directly on the can or on a label securely affixed to the can.

6. NOTES

6.1 Intended use. This product is intended for use with solvent in the dry cleaning of fabrics and clothing in dry cleaning equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(c) When bridge type handles are required (see 5.1.1).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

* 6.3 Carbon black. Supplied by L. Martin Company, Inc., 41 East 42nd Street, New York, NY 10017, has been found to be satisfactory.

6.4 Wool cloth. Cloth, wool, 100 percent virgin, unbleached. Approximately 47 ends and 41 picks per inch and approximately 6 ounces per square yard (203 g/m^2).

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* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the prior issue.

Custodians:

Army - GL
Air Force - 99

Preparing activity:

Army - GL

Project No. 7930-0244

Review activities:

Army - MD
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User activities:

Army - SM
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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